Received Order Analysis Engine (ROAE)

Analysis of the King Wen sequence including observations by Terence McKenna

Hexagram reference table

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Hexagram reference table
Each hexagram is a stack of 6 lines, either solid (1/yang) or broken (0/yin).
The 'binary' column encodes these 6 lines as bits (bottom line = rightmost bit).
Each hexagram is also composed of two 3-line trigrams: an upper and a lower.
There are 8 possible trigrams (Heaven, Earth, Thunder, Water, Mountain, Wind,
Fire, Lake), giving 8x8 = 64 possible hexagrams.
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Pos | Hex | Binary | Upper          | Lower          | Name
----|-----|--------|----------------|----------------|----
01  | ䷀   | 111111 | Qian Heaven    | Qian Heaven    | Heaven over Heaven
02  | ䷁   | 000000 | Kun  Earth     | Kun  Earth     | Earth over Earth
03  | ䷂   | 010001 | Kan  Water     | Zhen Thunder   | Water over Thunder
04  | ䷃   | 100010 | Gen  Mountain  | Kan  Water     | Mountain over Water
05  | ䷄   | 010111 | Kan  Water     | Qian Heaven    | Water over Heaven
06  | ䷅   | 111010 | Qian Heaven    | Kan  Water     | Heaven over Water
07  | ䷆   | 000010 | Kun  Earth     | Kan  Water     | Earth over Water
08  | ䷇   | 010000 | Kan  Water     | Kun  Earth     | Water over Earth
09  | ䷈   | 110111 | Xun  Wind      | Qian Heaven    | Wind over Heaven
10  | ䷉   | 111011 | Qian Heaven    | Dui  Lake      | Heaven over Lake
11  | ䷊   | 000111 | Kun  Earth     | Qian Heaven    | Earth over Heaven
12  | ䷋   | 111000 | Qian Heaven    | Kun  Earth     | Heaven over Earth
13  | ䷌   | 111101 | Qian Heaven    | Li   Fire      | Heaven over Fire
14  | ䷍   | 101111 | Li   Fire      | Qian Heaven    | Fire over Heaven
15  | ䷎   | 000100 | Kun  Earth     | Gen  Mountain  | Earth over Mountain
16  | ䷏   | 001000 | Zhen Thunder   | Kun  Earth     | Thunder over Earth
17  | ䷐   | 011001 | Dui  Lake      | Zhen Thunder   | Lake over Thunder
18  | ䷑   | 100110 | Gen  Mountain  | Xun  Wind      | Mountain over Wind
19  | ䷒   | 000011 | Kun  Earth     | Dui  Lake      | Earth over Lake
20  | ䷓   | 110000 | Xun  Wind      | Kun  Earth     | Wind over Earth
21  | ䷔   | 101001 | Li   Fire      | Zhen Thunder   | Fire over Thunder
22  | ䷕   | 100101 | Gen  Mountain  | Li   Fire      | Mountain over Fire
23  | ䷖   | 100000 | Gen  Mountain  | Kun  Earth     | Mountain over Earth
24  | ䷗   | 000001 | Kun  Earth     | Zhen Thunder   | Earth over Thunder
25  | ䷘   | 111001 | Qian Heaven    | Zhen Thunder   | Heaven over Thunder
26  | ䷙   | 100111 | Gen  Mountain  | Qian Heaven    | Mountain over Heaven
27  | ䷚   | 100001 | Gen  Mountain  | Zhen Thunder   | Mountain over Thunder
28  | ䷛   | 011110 | Dui  Lake      | Xun  Wind      | Lake over Wind
29  | ䷜   | 010010 | Kan  Water     | Kan  Water     | Water over Water
30  | ䷝   | 101101 | Li   Fire      | Li   Fire      | Fire over Fire
31  | ䷞   | 011100 | Dui  Lake      | Gen  Mountain  | Lake over Mountain
32  | ䷟   | 001110 | Zhen Thunder   | Xun  Wind      | Thunder over Wind
33  | ䷠   | 111100 | Qian Heaven    | Gen  Mountain  | Heaven over Mountain
34  | ䷡   | 001111 | Zhen Thunder   | Qian Heaven    | Thunder over Heaven
35  | ䷢   | 101000 | Li   Fire      | Kun  Earth     | Fire over Earth
36  | ䷣   | 000101 | Kun  Earth     | Li   Fire      | Earth over Fire
37  | ䷤   | 110101 | Xun  Wind      | Li   Fire      | Wind over Fire
38  | ䷥   | 101011 | Li   Fire      | Dui  Lake      | Fire over Lake
39  | ䷦   | 010100 | Kan  Water     | Gen  Mountain  | Water over Mountain
40  | ䷧   | 001010 | Zhen Thunder   | Kan  Water     | Thunder over Water
41  | ䷨   | 100011 | Gen  Mountain  | Dui  Lake      | Mountain over Lake
42  | ䷩   | 110001 | Xun  Wind      | Zhen Thunder   | Wind over Thunder
43  | ䷪   | 011111 | Dui  Lake      | Qian Heaven    | Lake over Heaven
44  | ䷫   | 111110 | Qian Heaven    | Xun  Wind      | Heaven over Wind
45  | ䷬   | 011000 | Dui  Lake      | Kun  Earth     | Lake over Earth
46  | ䷭   | 000110 | Kun  Earth     | Xun  Wind      | Earth over Wind
47  | ䷮   | 011010 | Dui  Lake      | Kan  Water     | Lake over Water
48  | ䷯   | 010110 | Kan  Water     | Xun  Wind      | Water over Wind
49  | ䷰   | 011101 | Dui  Lake      | Li   Fire      | Lake over Fire
50  | ䷱   | 101110 | Li   Fire      | Xun  Wind      | Fire over Wind
51  | ䷲   | 001001 | Zhen Thunder   | Zhen Thunder   | Thunder over Thunder
52  | ䷳   | 100100 | Gen  Mountain  | Gen  Mountain  | Mountain over Mountain
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53  | ䷴  | 110100 | Xun  Wind      | Gen  Mountain  | Wind over Mountain
54  | ䷵  | 001011 | Zhen Thunder   | Dui  Lake      | Thunder over Lake
55  | ䷶  | 001101 | Zhen Thunder   | Li   Fire      | Thunder over Fire
56  | ䷷  | 101100 | Li   Fire      | Gen  Mountain  | Fire over Mountain
57  | ䷸  | 110110 | Xun  Wind      | Xun  Wind      | Wind over Wind
58  | ䷹  | 011011 | Dui  Lake      | Dui  Lake      | Lake over Lake
59  | ䷺  | 110010 | Xun  Wind      | Kan  Water     | Wind over Water
60  | ䷻  | 010011 | Kan  Water     | Dui  Lake      | Water over Lake
61  | ䷼  | 110011 | Xun  Wind      | Dui  Lake      | Wind over Lake
62  | ䷽  | 001100 | Zhen Thunder   | Gen  Mountain  | Thunder over Mountain
63  | ䷾  | 010101 | Kan  Water     | Li   Fire      | Water over Fire
64  | ䷿  | 101010 | Li   Fire      | Kan  Water     | Fire over Water
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Reverse vs. Inverse pair analysis

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Reverse vs. Inverse pair analysis
The 64 hexagrams are traditionally grouped into 32 pairs (1-2, 3-4, ..., 63-64).
For each pair, we test two relationships:
  Reverse: flip the hexagram upside down (read the lines top-to-bottom instead of
           bottom-to-top). If the second hexagram upside-down equals the first,
           they are a reverse pair.
  Inverse: toggle every line (solid becomes broken, broken becomes solid).
           If toggling all lines of one hexagram gives the other, they are inverse.
Key finding: EVERY pair in the King Wen sequence is one or the other — none are
unrelated. This perfect pairing structure is unlikely to occur by chance.
---
01 ䷀ Inverse 02 ䷁
03 ䷂ Reverse 04 ䷃
05 ䷄ Reverse 06 ䷅
07 ䷆ Reverse 08 ䷇
09 ䷈ Reverse 10 ䷉
11 ䷊ Reverse 12 ䷋
13 ䷌ Reverse 14 ䷍
15 ䷎ Reverse 16 ䷏
17 ䷐ Reverse 18 ䷑
19 ䷒ Reverse 20 ䷓
21 ䷔ Reverse 22 ䷕
23 ䷖ Reverse 24 ䷗
25 ䷘ Reverse 26 ䷙
27 ䷚ Inverse 28 ䷛
29 ䷜ Inverse 30 ䷝
31 ䷞ Reverse 32 ䷟
33 ䷠ Reverse 34 ䷡
35 ䷢ Reverse 36 ䷣
37 ䷤ Reverse 38 ䷥
39 ䷦ Reverse 40 ䷧
41 ䷨ Reverse 42 ䷩
43 ䷪ Reverse 44 ䷫
45 ䷬ Reverse 46 ䷭
47 ䷮ Reverse 48 ䷯
49 ䷰ Reverse 50 ䷱
51 ䷲ Reverse 52 ䷳
53 ䷴ Reverse 54 ䷵
55 ䷶ Reverse 56 ䷷
57 ䷸ Reverse 58 ䷹
59 ䷺ Reverse 60 ䷻
61 ䷼ Inverse 62 ䷽
63 ䷾ Reverse 64 ䷿
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Reverse count pairs: 28/32 (87.5%)
Inverse count pairs: 04/32 (12.5%)
Neither count pairs: 00/32 (0.0%)
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First order of difference (the 'wave')

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First order of difference (the 'wave')
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For each consecutive pair of hexagrams, count how many of the 6 lines change.
This produces a sequence of values from 0 (identical) to 6 (every line flipped).
The resulting 'wave' is central to analysis of the King Wen sequence.
Key observation (McKenna, The Invisible Landscape, 1975): the value 5 never appears — no consecutive hexagrams
differ by exactly 5 lines. The value 0 also never appears (no duplicates).
---
01 ䷀ to 02 ䷁ difference 6 ******
02 ䷁ to 03 ䷂ difference 2 **
03 ䷂ to 04 ䷃ difference 4 ****
04 ䷃ to 05 ䷄ difference 4 ****
05 ䷄ to 06 ䷅ difference 4 ****
06 ䷅ to 07 ䷆ difference 3 ***
07 ䷆ to 08 ䷇ difference 2 **
08 ䷇ to 09 ䷈ difference 4 ****
09 ䷈ to 10 ䷉ difference 2 **
10 ䷉ to 11 ䷊ difference 4 ****
11 ䷊ to 12 ䷋ difference 6 ******
12 ䷋ to 13 ䷌ difference 2 **
13 ䷌ to 14 ䷍ difference 2 **
14 ䷍ to 15 ䷎ difference 4 ****
15 ䷎ to 16 ䷏ difference 2 **
16 ䷏ to 17 ䷐ difference 2 **
17 ䷐ to 18 ䷑ difference 6 ******
18 ䷑ to 19 ䷒ difference 3 ***
19 ䷒ to 20 ䷓ difference 4 ****
20 ䷓ to 21 ䷔ difference 3 ***
21 ䷔ to 22 ䷕ difference 2 **
22 ䷕ to 23 ䷖ difference 2 **
23 ䷖ to 24 ䷗ difference 2 **
24 ䷗ to 25 ䷘ difference 3 ***
25 ䷘ to 26 ䷙ difference 4 ****
26 ䷙ to 27 ䷚ difference 2 **
27 ䷚ to 28 ䷛ difference 6 ******
28 ䷛ to 29 ䷜ difference 2 **
29 ䷜ to 30 ䷝ difference 6 ******
30 ䷝ to 31 ䷞ difference 3 ***
31 ䷞ to 32 ䷟ difference 2 **
32 ䷟ to 33 ䷠ difference 3 ***
33 ䷠ to 34 ䷡ difference 4 ****
34 ䷡ to 35 ䷢ difference 4 ****
35 ䷢ to 36 ䷣ difference 4 ****
36 ䷣ to 37 ䷤ difference 2 **
37 ䷤ to 38 ䷥ difference 4 ****
38 ䷥ to 39 ䷦ difference 6 ******
39 ䷦ to 40 ䷧ difference 4 ****
40 ䷧ to 41 ䷨ difference 3 ***
41 ䷨ to 42 ䷩ difference 2 **
42 ䷩ to 43 ䷪ difference 4 ****
43 ䷪ to 44 ䷫ difference 2 **
44 ䷫ to 45 ䷬ difference 3 ***
45 ䷬ to 46 ䷭ difference 4 ****
46 ䷭ to 47 ䷮ difference 3 ***
47 ䷮ to 48 ䷯ difference 2 **
48 ䷯ to 49 ䷰ difference 3 ***
49 ䷰ to 50 ䷱ difference 4 ****
50 ䷱ to 51 ䷲ difference 4 ****
51 ䷲ to 52 ䷳ difference 4 ****
52 ䷳ to 53 ䷴ difference 1 *
53 ䷴ to 54 ䷵ difference 6 ******
54 ䷵ to 55 ䷶ difference 2 **
55 ䷶ to 56 ䷷ difference 2 **
56 ䷷ to 57 ䷸ difference 3 ***
57 ䷸ to 58 ䷹ difference 4 ****
58 ䷹ to 59 ䷺ difference 3 ***
59 ䷺ to 60 ䷻ difference 2 **
60 ䷻ to 61 ䷼ difference 1 *
61 ䷼ to 62 ䷽ difference 6 ******
62 ䷽ to 63 ䷾ difference 3 ***
63 ䷾ to 64 ䷿ difference 6 ******
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0 line changes total 0
1 line changes total 2
2 line changes total 20
3 line changes total 13
4 line changes total 19
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5 line changes total 0
6 line changes total 9

Spark line: █▂▅▅▅▃▂▅▂▅█▂▂▅▂▂█▃▅▃▂▂▂▃▅▂█▂█▃▂▃▅▅▅▂▅█▅▃▂▅▂▃▅▃▂▃▅▅▅▁█▂▂▃▅▃▂▁█▂█

--- Wald-Wolfowitz runs test ---
Tests whether the wave values are randomly ordered (vs. clustered or alternating).
Values split at median (3): 28 above, 22 below (13 excluded ties)
Observed runs: 33, expected: 25.6, Z = +2.13
Significant at 95% level (p = 0.033): the wave shows alternation.
Note: does not survive Bonferroni correction for 28 tests (threshold p < 0.0018).
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Difference wave bar chart

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Difference wave bar chart
A visual representation of the first-order difference wave. Each column shows
how many lines changed in that transition. The horizontal axis is the position
in the King Wen sequence (1-63), the vertical axis is the number of lines
changed (1-6). This makes the wave pattern easier to see than a list of numbers.
---
  6 |#         #     #         # #        #              #       # #|
  5 |#         #     #         # #        #              #       # #|
  4 |# ###  # ##  #  # #     # # #   ### ###  #  #   ### #   #   # #|
  3 |# #### # ##  #  ####   ## # ## #### #### # ### #### #  ###  ###|
  2 |################################################### ####### ###|
  1 |###############################################################|
    +---------------------------------------------------------------+
              1         2         3         4         5         6
              0         0         0         0         0         0
```

Trigram analysis

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Trigram analysis
Each hexagram is built from two trigrams (3-line figures): an upper and a lower.
There are 8 possible trigrams. This section counts how often each trigram appears
in each position, checks how often the upper or lower trigram changes between
consecutive hexagrams, and shows the full 8x8 transition matrix (which trigram
follows which). A uniform distribution (8 of each) confirms that all 64 possible
upper/lower combinations are used exactly once.
---
Trigram             Upper  Lower  Total
Kun  Earth            8      8     16
Zhen Thunder          8      8     16
Kan  Water            8      8     16
Dui  Lake             8      8     16
Gen  Mountain         8      8     16
Li   Fire             8      8     16
Xun  Wind             8      8     16
Qian Heaven           8      8     16

--- Consecutive trigram transitions ---
Upper trigram changes: 59/63 transitions (93.7%)
Lower trigram changes: 58/63 transitions (92.1%)

--- Upper trigram transition matrix ---
Rows = from, Columns = to
          Kun   Zhen  Kan   Dui   Gen   Li    Xun   Qian
Kun       0     1     2     1     0     0     2     2
Zhen      0     1     1     1     2     2     0     1
Kan       0     1     0     1     1     2     2     1
Dui       1     1     2     0     1     1     1     1
Gen       2     0     1     1     2     0     2     0
Li        2     1     1     1     1     0     1     0
Xun       0     2     1     2     0     2     0     1
Qian      3     1     0     1     1     1     0     1
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--- Lower trigram transition matrix ---
Rows = from, Columns = to
          Kun   Zhen  Kan   Dui   Gen   Li    Xun   Qian
Kun       0     4     0     0     0     2     1     1
Zhen      0     1     1     0     1     1     2     2
Kan       1     0     1     2     0     1     1     1
Dui       1     1     1     1     2     1     0     1
Gen       1     0     1     1     1     1     2     1
Li        1     0     1     1     2     1     1     1
Xun       1     1     2     2     1     1     0     0
Qian      3     1     1     1     1     0     1     0

--- Trigram change rates vs null (pair-preserving permutation test) ---
Null: 10,000 orderings preserving the pair structure (permute pairs + flip orientations),
the correct baseline per CRITIQUE.md. Statistic: upper/lower trigram change counts.
KW upper changes 59/63: percentile 47.3 (fraction of null <= KW)
KW lower changes 58/63: percentile 27.3

--- Pure (doubled-trigram) hexagram placement ---
The 8 pure hexagrams (upper == lower trigram). Lai Zhide (1525-1604, via Schulz 1982)
observed kan/li doubles closing both Classics; measured here against the same null.
Pure hexagram positions (1-based): [1, 2, 29, 30, 51, 52, 57, 58]
Pure hexagrams at Classic ends (positions 1,2,29,30,63,64): 4 of a possible 6
Null P(>= KW's 4) = 0.0336  (pair-preserving null; note KW's C4 fixes 1,2 by
definition, so interpret against the constrained baseline)

--- Nuclear trigram structure ---
Nuclear hexagram = lines 2-4 (lower nuclear) + 3-5 (upper nuclear). Classical fact
(commentary tradition): iterating the nuclear map sends all 64 hexagrams to 16, then
to a 4-element terminal attractor {0, 21, 42, 63}. CORRECTED 2026-08-01: an earlier
version called these '4 fixed points', which is false and self-contradicting -- it
then called two of them 'alternators'. There are exactly TWO fixed points, 0 and 63;
21 and 42 form a 2-cycle (nuclear(21) = 42, nuclear(42) = 21). The 16/4/4 counts below
are unaffected.
Distinct after 1 application: 16; after 2: 4; after 3: 4 -> [0, 21, 42, 63]
KW nuclear-hexagram changes along the sequence: 58/63

--- Jing Fang Eight-Palaces rank correlation ---
Spearman rank correlation between KW positions and Jing Fang's palace ordering
(the trigram-generated classical alternative; ordering as in solve.c --null-historical).
Spearman rho = 0.1384; null P(|rho| >= observed) = 0.1170 (pair-preserving null)

--- Symmetry group vs the trigram decomposition (ROAE SYMMETRY_SEARCH theorem) ---
The C1-C5 symmetry group B3 (order 48) acts on LINE positions; reversal maps the upper
trigram to the reversed lower trigram, so most of the group does NOT preserve the
upper/lower split. Preserving elements = permutations acting within {0,1,2} and {3,4,5}
blocks (or swapping them wholesale) that also centralize rev.
Split-respecting subgroup order: 12 of 48 (computed over the rev-centralizer)

--- Methodological note ---
With ~1 expected observation per cell, no goodness-of-fit test (e.g.,
chi-square) has sufficient power to detect deviations from uniform
transitions. The matrices are descriptive only.
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Nuclear hexagram analysis

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Nuclear hexagram analysis
Every hexagram contains a hidden 'nuclear' hexagram inside it, formed by
extracting the 4 middle lines (lines 2-3-4-5). Lines 2-3-4 form the lower
nuclear trigram, and lines 3-4-5 form the upper nuclear trigram (sharing lines
3 and 4). This creates a chain: each hexagram points to its nuclear hexagram,
which points to its own nuclear hexagram, and so on until the chain loops.
These chains reveal deep structural relationships between hexagrams that are
not obvious from the surface arrangement.
---
01 ䷀ nuclear: 01 ䷀  chain: 01 ䷀ -> 01 ䷀
02 ䷁ nuclear: 02 ䷁  chain: 02 ䷁ -> 02 ䷁
03 ䷂ nuclear: 23 ䷖  chain: 03 ䷂ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
04 ䷃ nuclear: 24 ䷗  chain: 04 ䷃ -> 24 ䷗ -> 02 ䷁ -> 02 ䷁
05 ䷄ nuclear: 38 ䷥  chain: 05 ䷄ -> 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
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06 ䷅ nuclear: 37 ䷤  chain: 06 ䷅ -> 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
07 ䷆ nuclear: 24 ䷗  chain: 07 ䷆ -> 24 ䷗ -> 02 ䷁ -> 02 ䷁
08 ䷇ nuclear: 23 ䷖  chain: 08 ䷇ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
09 ䷈ nuclear: 38 ䷥  chain: 09 ䷈ -> 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
10 ䷉ nuclear: 37 ䷤  chain: 10 ䷉ -> 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
11 ䷊ nuclear: 54 ䷵  chain: 11 ䷊ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
12 ䷋ nuclear: 53 ䷴  chain: 12 ䷋ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
13 ䷌ nuclear: 44 ䷫  chain: 13 ䷌ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
14 ䷍ nuclear: 43 ䷪  chain: 14 ䷍ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
15 ䷎ nuclear: 40 ䷧  chain: 15 ䷎ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
16 ䷏ nuclear: 39 ䷦  chain: 16 ䷏ -> 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
17 ䷐ nuclear: 53 ䷴  chain: 17 ䷐ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
18 ䷑ nuclear: 54 ䷵  chain: 18 ䷑ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
19 ䷒ nuclear: 24 ䷗  chain: 19 ䷒ -> 24 ䷗ -> 02 ䷁ -> 02 ䷁
20 ䷓ nuclear: 23 ䷖  chain: 20 ䷓ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
21 ䷔ nuclear: 39 ䷦  chain: 21 ䷔ -> 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
22 ䷕ nuclear: 40 ䷧  chain: 22 ䷕ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
23 ䷖ nuclear: 02 ䷁  chain: 23 ䷖ -> 02 ䷁ -> 02 ䷁
24 ䷗ nuclear: 02 ䷁  chain: 24 ䷗ -> 02 ䷁ -> 02 ䷁
25 ䷘ nuclear: 53 ䷴  chain: 25 ䷘ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
26 ䷙ nuclear: 54 ䷵  chain: 26 ䷙ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
27 ䷚ nuclear: 02 ䷁  chain: 27 ䷚ -> 02 ䷁ -> 02 ䷁
28 ䷛ nuclear: 01 ䷀  chain: 28 ䷛ -> 01 ䷀ -> 01 ䷀
29 ䷜ nuclear: 27 ䷚  chain: 29 ䷜ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
30 ䷝ nuclear: 28 ䷛  chain: 30 ䷝ -> 28 ䷛ -> 01 ䷀ -> 01 ䷀
31 ䷞ nuclear: 44 ䷫  chain: 31 ䷞ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
32 ䷟ nuclear: 43 ䷪  chain: 32 ䷟ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
33 ䷠ nuclear: 44 ䷫  chain: 33 ䷠ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
34 ䷡ nuclear: 43 ䷪  chain: 34 ䷡ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
35 ䷢ nuclear: 39 ䷦  chain: 35 ䷢ -> 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
36 ䷣ nuclear: 40 ䷧  chain: 36 ䷣ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
37 ䷤ nuclear: 64 ䷿  chain: 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
38 ䷥ nuclear: 63 ䷾  chain: 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
39 ䷦ nuclear: 64 ䷿  chain: 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
40 ䷧ nuclear: 63 ䷾  chain: 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
41 ䷨ nuclear: 24 ䷗  chain: 41 ䷨ -> 24 ䷗ -> 02 ䷁ -> 02 ䷁
42 ䷩ nuclear: 23 ䷖  chain: 42 ䷩ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
43 ䷪ nuclear: 01 ䷀  chain: 43 ䷪ -> 01 ䷀ -> 01 ䷀
44 ䷫ nuclear: 01 ䷀  chain: 44 ䷫ -> 01 ䷀ -> 01 ䷀
45 ䷬ nuclear: 53 ䷴  chain: 45 ䷬ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
46 ䷭ nuclear: 54 ䷵  chain: 46 ䷭ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
47 ䷮ nuclear: 37 ䷤  chain: 47 ䷮ -> 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
48 ䷯ nuclear: 38 ䷥  chain: 48 ䷯ -> 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
49 ䷰ nuclear: 44 ䷫  chain: 49 ䷰ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
50 ䷱ nuclear: 43 ䷪  chain: 50 ䷱ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
51 ䷲ nuclear: 39 ䷦  chain: 51 ䷲ -> 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
52 ䷳ nuclear: 40 ䷧  chain: 52 ䷳ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
53 ䷴ nuclear: 64 ䷿  chain: 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
54 ䷵ nuclear: 63 ䷾  chain: 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
55 ䷶ nuclear: 28 ䷛  chain: 55 ䷶ -> 28 ䷛ -> 01 ䷀ -> 01 ䷀
56 ䷷ nuclear: 28 ䷛  chain: 56 ䷷ -> 28 ䷛ -> 01 ䷀ -> 01 ䷀
57 ䷸ nuclear: 38 ䷥  chain: 57 ䷸ -> 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
58 ䷹ nuclear: 37 ䷤  chain: 58 ䷹ -> 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
59 ䷺ nuclear: 27 ䷚  chain: 59 ䷺ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
60 ䷻ nuclear: 27 ䷚  chain: 60 ䷻ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
61 ䷼ nuclear: 27 ䷚  chain: 61 ䷼ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
62 ䷽ nuclear: 28 ䷛  chain: 62 ䷽ -> 28 ䷛ -> 01 ䷀ -> 01 ䷀
63 ䷾ nuclear: 64 ䷿  chain: 63 ䷾ -> 64 ䷿ -> 63 ䷾
64 ䷿ nuclear: 63 ䷾  chain: 64 ䷿ -> 63 ䷾ -> 64 ䷿

--- Nuclear hexagram frequency ---
How often each hexagram appears as another's nuclear hexagram
01 ䷀ Heaven over Heaven            appears 4 times
02 ䷁ Earth over Earth              appears 4 times
23 ䷖ Mountain over Earth           appears 4 times
24 ䷗ Earth over Thunder            appears 4 times
27 ䷚ Mountain over Thunder         appears 4 times
28 ䷛ Lake over Wind                appears 4 times
37 ䷤ Wind over Fire                appears 4 times
38 ䷥ Fire over Lake                appears 4 times
39 ䷦ Water over Mountain           appears 4 times
40 ䷧ Thunder over Water            appears 4 times
43 ䷪ Lake over Heaven              appears 4 times
44 ䷫ Heaven over Wind              appears 4 times
53 ䷴ Wind over Mountain            appears 4 times
```

```
54 ䷵ Thunder over Lake              appears 4 times
63 ䷾ Water over Fire                appears 4 times
64 ䷿ Fire over Water                appears 4 times

--- Null model note ---
Nuclear hexagram derivation is a fixed function of binary value (lines 2-5),
independent of the King Wen ordering. The chain structure, cycle lengths,
and frequency distribution above are identical for ANY ordering of the 64
hexagrams. These properties reflect the 6-bit binary system, not the
King Wen sequence specifically.
```

Line change positional analysis

```
---
Line change positional analysis
Each hexagram has 6 lines (positions 1-6, bottom to top). When consecutive
hexagrams differ, which specific lines are changing? If the King Wen sequence
were random, each line position would change about 50% of the time. Significant
deviation from 50% would suggest the ordering favors changing certain positions.
The co-change matrix shows which pairs of lines tend to flip together in the
same transition — revealing correlated line movements.
---
Line 1: 37/63 ( 58.7%)  #####################################
Line 2: 32/63 ( 50.8%)  ################################
Line 3: 35/63 ( 55.6%)  ###################################
Line 4: 34/63 ( 54.0%)  ##################################
Line 5: 35/63 ( 55.6%)  ###################################
Line 6: 38/63 ( 60.3%)  ######################################

--- Line co-change frequency ---
How often pairs of lines change together in the same transition
      L1   L2   L3   L4   L5   L6
L1    .    18   20   19   19   30
L2    .    .    20   17   23   20
L3    .    .    .    25   16   21
L4    .    .    .    .    20   20
L5    .    .    .    .    .    20
L6    .    .    .    .    .    .
```

Complement distance analysis

```
---
Complement distance analysis
Every hexagram has an 'opposite' (complement) formed by toggling all 6 lines
(solid becomes broken, broken becomes solid). For example, hexagram 1 (all
solid) and hexagram 2 (all broken) are complements. This section finds where
each hexagram's complement sits in the King Wen sequence and how far apart they
are. If complements sit nearer each other than a null model predicts, opposition
is an organizing feature of the sequence's structure; if farther, complements
may serve as bookends for larger structural sections.
---
01 ䷀ <-> 02 ䷁  distance:  1  Heaven over Heaven <-> Earth over Earth
02 ䷁ <-> 01 ䷀  distance:  1  Earth over Earth <-> Heaven over Heaven
03 ䷂ <-> 50 ䷱  distance: 47  Water over Thunder <-> Fire over Wind
04 ䷃ <-> 49 ䷰  distance: 45  Mountain over Water <-> Lake over Fire
05 ䷄ <-> 35 ䷢  distance: 30  Water over Heaven <-> Fire over Earth
06 ䷅ <-> 36 ䷣  distance: 30  Heaven over Water <-> Earth over Fire
07 ䷆ <-> 13 ䷌  distance:  6  Earth over Water <-> Heaven over Fire
08 ䷇ <-> 14 ䷍  distance:  6  Water over Earth <-> Fire over Heaven
09 ䷈ <-> 16 ䷏  distance:  7  Wind over Heaven <-> Thunder over Earth
10 ䷉ <-> 15 ䷎  distance:  5  Heaven over Lake <-> Earth over Mountain
11 ䷊ <-> 12 ䷋  distance:  1  Earth over Heaven <-> Heaven over Earth
12 ䷋ <-> 11 ䷊  distance:  1  Heaven over Earth <-> Earth over Heaven
13 ䷌ <-> 07 ䷆  distance:  6  Heaven over Fire <-> Earth over Water
14 ䷍ <-> 08 ䷇  distance:  6  Fire over Heaven <-> Water over Earth
15 ䷎ <-> 10 ䷉  distance:  5  Earth over Mountain <-> Heaven over Lake
16 ䷏ <-> 09 ䷈  distance:  7  Thunder over Earth <-> Wind over Heaven
17 ䷐ <-> 18 ䷑  distance:  1  Lake over Thunder <-> Mountain over Wind
```

```
18 ䷑ <-> 17 ䷐  distance:  1  Mountain over Wind <-> Lake over Thunder
19 ䷒ <-> 33 ䷠  distance: 14  Earth over Lake <-> Heaven over Mountain
20 ䷓ <-> 34 ䷡  distance: 14  Wind over Earth <-> Thunder over Heaven
21 ䷔ <-> 48 ䷯  distance: 27  Fire over Thunder <-> Water over Wind
22 ䷕ <-> 47 ䷮  distance: 25  Mountain over Fire <-> Lake over Water
23 ䷖ <-> 43 ䷪  distance: 20  Mountain over Earth <-> Lake over Heaven
24 ䷗ <-> 44 ䷫  distance: 20  Earth over Thunder <-> Heaven over Wind
25 ䷘ <-> 46 ䷭  distance: 21  Heaven over Thunder <-> Earth over Wind
26 ䷙ <-> 45 ䷬  distance: 19  Mountain over Heaven <-> Lake over Earth
27 ䷚ <-> 28 ䷛  distance:  1  Mountain over Thunder <-> Lake over Wind
28 ䷛ <-> 27 ䷚  distance:  1  Lake over Wind <-> Mountain over Thunder
29 ䷜ <-> 30 ䷝  distance:  1  Water over Water <-> Fire over Fire
30 ䷝ <-> 29 ䷜  distance:  1  Fire over Fire <-> Water over Water
31 ䷞ <-> 41 ䷨  distance: 10  Lake over Mountain <-> Mountain over Lake
32 ䷟ <-> 42 ䷩  distance: 10  Thunder over Wind <-> Wind over Thunder
33 ䷠ <-> 19 ䷒  distance: 14  Heaven over Mountain <-> Earth over Lake
34 ䷡ <-> 20 ䷓  distance: 14  Thunder over Heaven <-> Wind over Earth
35 ䷢ <-> 05 ䷄  distance: 30  Fire over Earth <-> Water over Heaven
36 ䷣ <-> 06 ䷅  distance: 30  Earth over Fire <-> Heaven over Water
37 ䷤ <-> 40 ䷧  distance:  3  Wind over Fire <-> Thunder over Water
38 ䷥ <-> 39 ䷦  distance:  1  Fire over Lake <-> Water over Mountain
39 ䷦ <-> 38 ䷥  distance:  1  Water over Mountain <-> Fire over Lake
40 ䷧ <-> 37 ䷤  distance:  3  Thunder over Water <-> Wind over Fire
41 ䷨ <-> 31 ䷞  distance: 10  Mountain over Lake <-> Lake over Mountain
42 ䷩ <-> 32 ䷟  distance: 10  Wind over Thunder <-> Thunder over Wind
43 ䷪ <-> 23 ䷖  distance: 20  Lake over Heaven <-> Mountain over Earth
44 ䷫ <-> 24 ䷗  distance: 20  Heaven over Wind <-> Earth over Thunder
45 ䷬ <-> 26 ䷙  distance: 19  Lake over Earth <-> Mountain over Heaven
46 ䷭ <-> 25 ䷘  distance: 21  Earth over Wind <-> Heaven over Thunder
47 ䷮ <-> 22 ䷕  distance: 25  Lake over Water <-> Mountain over Fire
48 ䷯ <-> 21 ䷔  distance: 27  Water over Wind <-> Fire over Thunder
49 ䷰ <-> 04 ䷃  distance: 45  Lake over Fire <-> Mountain over Water
50 ䷱ <-> 03 ䷂  distance: 47  Fire over Wind <-> Water over Thunder
51 ䷲ <-> 57 ䷸  distance:  6  Thunder over Thunder <-> Wind over Wind
52 ䷳ <-> 58 ䷹  distance:  6  Mountain over Mountain <-> Lake over Lake
53 ䷴ <-> 54 ䷵  distance:  1  Wind over Mountain <-> Thunder over Lake
54 ䷵ <-> 53 ䷴  distance:  1  Thunder over Lake <-> Wind over Mountain
55 ䷶ <-> 59 ䷺  distance:  4  Thunder over Fire <-> Wind over Water
56 ䷷ <-> 60 ䷻  distance:  4  Fire over Mountain <-> Water over Lake
57 ䷸ <-> 51 ䷲  distance:  6  Wind over Wind <-> Thunder over Thunder
58 ䷹ <-> 52 ䷳  distance:  6  Lake over Lake <-> Mountain over Mountain
59 ䷺ <-> 55 ䷶  distance:  4  Wind over Water <-> Thunder over Fire
60 ䷻ <-> 56 ䷷  distance:  4  Water over Lake <-> Fire over Mountain
61 ䷼ <-> 62 ䷽  distance:  1  Wind over Lake <-> Thunder over Mountain
62 ䷽ <-> 61 ䷼  distance:  1  Thunder over Mountain <-> Wind over Lake
63 ䷾ <-> 64 ䷿  distance:  1  Water over Fire <-> Fire over Water
64 ䷿ <-> 63 ䷾  distance:  1  Fire over Water <-> Water over Fire

--- Complement distance statistics ---
Mean distance:   12.1
Median distance: 6
Min distance:    1 (adjacent = pair is an inverse pair)
Max distance:    47

--- Complement distance null model ---
Random mean complement distance (over 10000 shuffles): 21.7
King Wen mean complement distance: 12.1
King Wen percentile vs random: 0.0%
Complements are significantly closer together than chance would predict:
the ordering keeps complements unusually near one another.
```

Palindrome analysis of the difference wave

```
---
Palindrome analysis of the difference wave
A palindrome reads the same forwards and backwards (like 2,4,6,4,2). Palindromic
runs in the difference wave are mirror symmetry — sections where the pattern of
change rises and falls in a balanced way. Longer palindromes are less likely to
occur by chance, so counts and lengths are measured against null models below.
---
Found 27 palindromic subsequences (showing longest 20)
```

```
Length  7 at positions 11-17: [6,2,2,4,2,2,6] █▁▁▄▁▁█
Length  7 at positions 19-25: [4,3,2,2,2,3,4] ▄▂▁▁▁▂▄
Length  7 at positions 39-45: [4,3,2,4,2,3,4] ▄▂▁▄▁▂▄
Length  5 at positions 12-16: [2,2,4,2,2] ▁▁▄▁▁
Length  5 at positions 20-24: [3,2,2,2,3] ▂▁▁▁▂
Length  5 at positions 40-44: [3,2,4,2,3] ▂▁▄▁▂
Length  5 at positions 43-47: [2,3,4,3,2] ▁▂▄▂▁
Length  5 at positions 45-49: [4,3,2,3,4] ▄▂▁▂▄
Length  5 at positions 55-59: [2,3,4,3,2] ▁▂▄▂▁
Length  3 at positions 03-05: [4,4,4] ▄▄▄
Length  3 at positions 07-09: [2,4,2] ▁▄▁
Length  3 at positions 08-10: [4,2,4] ▄▁▄
Length  3 at positions 13-15: [2,4,2] ▁▄▁
Length  3 at positions 18-20: [3,4,3] ▂▄▂
Length  3 at positions 21-23: [2,2,2] ▁▁▁
Length  3 at positions 26-28: [2,6,2] ▁█▁
Length  3 at positions 27-29: [6,2,6] █▁█
Length  3 at positions 30-32: [3,2,3] ▂▁▂
Length  3 at positions 33-35: [4,4,4] ▄▄▄
Length  3 at positions 35-37: [4,2,4] ▄▁▄

--- Palindrome count by length ---
Length  7: 3 found
Length  5: 6 found
Length  3: 18 found

--- Palindrome null model ---
How many palindromes of length >= 3 do random permutations produce?
King Wen palindromes: 27, longest: 7
Random mean palindromes: 22.1, mean longest: 6.9
King Wen palindrome count percentile: 79.1%
King Wen longest palindrome percentile: 41.0%

--- Pair-constrained palindrome null model ---
The unconstrained comparison shuffles all 64 hexagrams freely. A fairer
comparison preserves the pair structure: each pair of hexagrams stays
adjacent, only the pair order and within-pair orientation are randomized.
King Wen palindromes: 27, longest: 7
Pair-constrained mean palindromes: 27.6, mean longest: 8.4
King Wen palindrome count percentile (pair-constrained): 49.3%
King Wen longest palindrome percentile (pair-constrained): 13.2%
```

Upper Canon (1-30) vs. Lower Canon (31-64) comparison

```
---
Upper Canon (1-30) vs. Lower Canon (31-64) comparison
The I Ching is traditionally divided into two books: the Upper Canon (hexagrams
1-30) and Lower Canon (hexagrams 31-64). This division is ancient and may reflect
different authorship, time periods, or thematic groupings. This section compares
the mathematical properties of each half: do they have similar difference wave
patterns? Similar yin/yang balance? Does the 'no 5-line transition' property hold
in each half independently, or only in the whole?
---
Metric                         Upper (1-30)  Lower (31-64)
------------------------------ ------------ --------------
Hexagrams                                30            34
Transitions                              29            33
Mean line changes                      3.38          3.33
Yang lines (total)                       86           106
Yin lines (total)                        94            98

Difference distribution:       Upper (1-30)  Lower (31-64)
0 line changes                             0             0
1 line changes                             0             2
2 line changes                            12             8
3 line changes                             4             8
4 line changes                             8            11
5 line changes                             0             0
6 line changes                             5             4
```

```
Upper Canon wave: █▂▅▅▅▃▂▅▂▅█▂▂▅▂▂█▃▅▃▂▂▂▃▅▂█▂█
Lower Canon wave: ▂▃▅▅▅▂▅█▅▃▂▅▂▃▅▃▂▃▅▅▅▁█▂▂▃▅▃▂▁█▃█

5-line transitions in Upper Canon: 0
5-line transitions in Lower Canon: 0

--- Canon split null model ---
Permutation test: is the King Wen split at position 30 special, or would
any random split of the 64-hexagram sequence show a similar gap in mean
line-change differences between the two halves?
King Wen |upper_mean - lower_mean|: 0.0460
Random permutations with gap >= King Wen: 8807/10000
King Wen gap percentile: 11.9%
The King Wen canon split does not produce a statistically significant
mean-difference gap. The observed difference between upper and lower
halves is consistent with what random shuffles produce.
```

Autocorrelation of the difference wave

```
---
Autocorrelation of the difference wave
Autocorrelation measures whether the wave is correlated with a shifted copy of
itself. A peak at lag N means the pattern tends to repeat every N steps. A value
near +1 means strong repetition, near -1 means strong alternation, and near 0
means no relationship. If the King Wen ordering contains hidden periodic
structure, it will show up as peaks in this analysis.
Caveat: with only N=63 data points, the 95% noise threshold is +/-0.25.
Values within that band are indistinguishable from random noise.
---
Mean difference: 3.349
Variance: 1.910
95% noise threshold: +/-0.247 (values inside this are not significant)

Lag    Autocorrelation  Sig?  Visualization
---    ---------------  ----  -------------
0               1.0000        [            :  |  :          #]
1              -0.2469        [            #  |  :           ]
2              -0.0098        [            :  |  :           ]
3              -0.0491        [            :  |  :           ]
4              -0.0447        [            :  |  :           ]
5              -0.0457        [            :  |  :           ]
6              -0.0218        [            :  |  :           ]
7              -0.0644        [            :  |  :           ]
8               0.1204        [            :  |# :           ]
9              -0.1275        [            : #|  :           ]
10              0.1375        [            :  | #:           ]
11             -0.0290        [            :  |  :           ]
12             -0.0221        [            :  |  :           ]
13              0.0850        [            :  |# :           ]
14             -0.1596        [            :# |  :           ]
15             -0.0110        [            :  |  :           ]
16              0.1015        [            :  |# :           ]
17             -0.1514        [            :# |  :           ]
18              0.0055        [            :  |  :           ]
19              0.0851        [            :  |# :           ]
20             -0.1403        [            :# |  :           ]
21              0.0274        [            :  |  :           ]
22             -0.1315        [            : #|  :           ]
23             -0.0137        [            :  |  :           ]
24              0.2263        [            :  |  #           ]
25             -0.1595        [            :# |  :           ]
26              0.1112        [            :  |# :           ]
27             -0.1109        [            : #|  :           ]
28              0.1649        [            :  | #:           ]
29              0.0093        [            :  |  :           ]
30             -0.0969        [            : #|  :           ]
31             -0.0480        [            :  |  :           ]

Significant lags (outside noise band): 0/31
(Expected false positives at 95% threshold with 31 lags: ~1.6)
No significant autocorrelation detected — consistent with no hidden periodicity,
```

```
but note that N=63 provides limited statistical power to detect weak periodicity.
```

Spectral analysis (Discrete Fourier Transform)

```
---
Spectral analysis (Discrete Fourier Transform)
The DFT decomposes the difference wave into frequency components, like splitting
white light into a rainbow. Each frequency represents a periodic pattern in the
wave. A strong peak at frequency F means the wave has a repeating cycle every
63/F transitions. If the King Wen ordering has hidden periodic structure, the
spectrum will show clear peaks rather than flat noise.
Caveat: with only 63 samples, each of the 31 frequency bins has wide
uncertainty. A flat spectrum may indicate no periodicity, or insufficient data.
---
Number of samples: 63
Mean (DC component): 3.349
White noise floor: 0.1741 (magnitudes near this are not significant)

Freq   Period  Magnitude  Sig?  Spectrum
----   ------  ---------  ----  --------
   1     63.0     0.0224        ##
   2     31.5     0.1823        #################
   3     21.0     0.0145        #
   4     15.8     0.1359        ############
   5     12.6     0.2021        ##################
   6     10.5     0.1157        ##########
   7      9.0     0.2399        ######################
   8      7.9     0.1092        ##########
   9      7.0     0.0553        #####
  10      6.3     0.1630        ###############
  11      5.7     0.2371        ######################
  12      5.2     0.0473        ####
  13      4.8     0.2326        #####################
  14      4.5     0.0331        ###
  15      4.2     0.1145        ##########
  16      3.9     0.2204        ####################
  17      3.7     0.1451        #############
  18      3.5     0.1975        ##################
  19      3.3     0.2197        ####################
  20      3.1     0.2166        ####################
  21      3.0     0.1611        ###############
  22      2.9     0.0222        ##
  23      2.7     0.1210        ###########
  24      2.6     0.4266     *  ########################################
  25      2.5     0.0306        ##
  26      2.4     0.2297        #####################
  27      2.3     0.0795        #######
  28      2.2     0.0639        #####
  29      2.2     0.2114        ###################
  30      2.1     0.2129        ###################
  31      2.0     0.2033        ###################

Frequencies above 2x noise floor: 1/31
(The 2x threshold is ad hoc. A proper test would use Fisher's g-statistic or
Bonferroni correction across frequency bins. With only 63 samples, even real
periodicity may not rise above the noise floor.)
```

Markov chain analysis

```
---
Markov chain analysis
If we treat each difference value as a 'state', we can ask: does the current
difference predict the next one? In a random sequence, knowing that the last
transition changed 2 lines tells you nothing about the next transition. But if
certain patterns like '6 is always followed by 2' emerge, it reveals hidden
rules in the King Wen ordering. The matrix shows the probability of each
transition between difference values.
---
```

```
Transition probability matrix (row = from, column = to)
        -> 1    -> 2    -> 3    -> 4    -> 6
  1     0.00    0.00    0.00    0.00    1.00    (n=2)
  2     0.05    0.25    0.25    0.30    0.15    (n=20)
  3     0.00    0.46    0.00    0.46    0.08    (n=13)
  4     0.05    0.26    0.26    0.32    0.11    (n=19)
  6     0.00    0.50    0.38    0.12    0.00    (n=8)

--- Notable transition patterns ---
(Caveat: rows with small n are unreliable — patterns based on <5 observations
are anecdotes, not statistically meaningful.)
  3 -> 2: 46% (6/13 times)
  3 -> 4: 46% (6/13 times)

--- Permutation test ---
Is the transition matrix more concentrated than random orderings produce?
King Wen matrix concentration: 2.3303
Random orderings equally or more concentrated: 2186/5000 (43.7%)
```

Gray code comparison

```
---
Gray code comparison
A Gray code is an ordering of binary values where each consecutive pair differs
by exactly 1 bit. For 6-bit hexagrams, a Gray code path would change exactly 1
line at every step — the smoothest possible sequence. The King Wen sequence does
NOT follow a Gray code (most transitions change 2-6 lines). This comparison
shows how much 'rougher' King Wen is than the theoretical minimum, suggesting
the ordering prioritizes something other than smooth transitions.
---
Metric                               Gray Code   King Wen
----------------------------------- ---------- ----------
Total path length                           63        211
Mean change per transition                1.00       3.35
Max single transition                        1          6
Min single transition                        1          1
King Wen / Gray Code ratio                           3.35x

Difference distribution:             Gray Code   King Wen
  1 line changes                             63          2
  2 line changes                              0         20
  3 line changes                              0         13
  4 line changes                              0         19
  6 line changes                              0          9

Gray code wave:
King Wen wave:
```

```
--- Methodological note ---
This comparison is descriptive. The Gray code ratio (King Wen path / Gray
code path) is not a statistical test. For significance testing of King Wen's
path length against appropriate null models, see --path.
```

Symmetry group analysis (XOR algebra)

```
---
Symmetry group analysis (XOR algebra)
The 64 hexagrams form a mathematical group under XOR: combining any two hexagrams
by toggling their differing lines always produces another valid hexagram. This is
like mixing colors — every combination is itself a color. Under this algebra,
certain sets of hexagrams form 'subgroups' (closed sets where combining any two
members stays in the set). This section checks whether the King Wen pairs, inverse
pairs, and the canonical divisions correspond to meaningful algebraic subgroups.
---
XOR identity element: 0b000000 (hexagram 2, all broken)

--- Fixed points under complement (XOR 0b111111) ---
Hexagrams that are their own complement do not exist (since XOR with
```

```
0b111111 always flips all bits, no 6-bit value equals its own complement).

--- King Wen pair XOR products ---
For each pair, XOR the two hexagrams to see what 'difference element' they produce.
If many pairs produce the same XOR product, the pairing has algebraic regularity.

XOR Product      Hexagram  Pair count  Pairs
  001100   62 ䷽       4      9-10, 15-16, 21-22, 47-48
  010010   29 ䷜       4      7-8, 13-14, 31-32, 41-42
  011110   28 ䷛       4      25-26, 37-38, 39-40, 45-46
  100001   27 ䷚       4      23-24, 43-44, 55-56, 59-60
  101101   30 ䷝       4      5-6, 35-36, 51-52, 57-58
  110011   61 ䷼       4      3-4, 19-20, 33-34, 49-50
  111111   01 ䷀       8      1-2, 11-12, 17-18, 27-28, 29-30, 53-54, 61-62, 63-64

Unique XOR products: 7 (out of 32 pairs)
If this number is small, the pairs share algebraic structure.

--- Inverse pair XOR closure test ---
Inverse pair hexagrams (16 total): 01, 02, 11, 12, 17, 18, 27, 28, 29, 30, 53, 54, 61, 62, 63, 64
Closed under XOR: Yes — forms a subgroup
```

Alternative sequence comparison

```
---
Alternative sequence comparison
The King Wen ordering is not the only way to arrange 64 hexagrams. The Fu Xi
(binary) sequence orders them by numerical value (0-63), which is mathematically
natural but has no traditional significance. The Mawangdui sequence was found on
silk manuscripts in a 168 BCE tomb and may represent an independent tradition.
Comparing the same analyses across orderings reveals what is unique to King Wen.
---
Metric                                King Wen   Fu Xi (binary)        Mawangdui
------------------------- ---------------- ---------------- ----------------
Total path length                          211              120              141
Mean change                               3.35             1.90             2.24
1-line transitions                           2               32               21
2-line transitions                          20               16               10
3-line transitions                          13                8               29
4-line transitions                          19                4                2
5-line transitions                           0                2                1
6-line transitions                           9                1                0

Waves:
  King Wen
  Fu Xi (binary)
  Mawangdui
```

```
--- What makes King Wen unique ---
Zero 5-line transitions: King Wen=0, Fu Xi=2, Mawangdui=1
Zero 0-line transitions: King Wen=0, Fu Xi=0, Mawangdui=0
```

Windowed entropy analysis

```
---
Windowed entropy analysis
Instead of one entropy value for the whole wave, we slide a window across it
and compute entropy locally. This reveals WHERE in the sequence structure
concentrates (low entropy = predictable) vs. dissolves (high entropy = varied).
Dips in the curve mark regions with repetitive transition patterns.
---
Window size: 15
Center  Entropy    Visualization
------  -------    -------------
     8   1.7056   ########################
     9   1.5628   ######################
    10   1.7056   ########################
    11   1.8323   ##########################
```

```
    12   1.8323   ##########################
    13   1.8892   ##########################
    14   1.7968   #########################
    15   1.7968   #########################
    16   1.7232   ########################
    17   1.8295   ##########################
    18   1.8295   ##########################
    19   1.6729   #######################
    20   1.8295   ##########################
    21   1.8295   ##########################
    22   1.8295   ##########################
    23   1.8892   ##########################
    24   1.8892   ##########################
    25   1.8323   ##########################
    26   1.8892   ##########################
    27   1.8892   ##########################
    28   1.8892   ##########################
    29   1.8892   ##########################
    30   1.9086   ###########################
    31   1.9656   ############################
    32   1.8892   ##########################
    33   1.9656   ############################
    34   1.9656   ############################
    35   1.8892   ##########################
    36   1.8892   ##########################
    37   1.8062   #########################
    38   1.7465   ########################
    39   1.7465   ########################
    40   1.7465   ########################
    41   1.8062   #########################
    42   1.8062   #########################
    43   1.8062   #########################
    44   1.7465   ########################
    45   2.0226   ############################
    46   2.0226   ############################
    47   2.0662   #############################
    48   2.0419   #############################
    49   2.0662   #############################
    50   2.0662   #############################
    51   2.0419   #############################
    52   2.0662   #############################
    53   2.1736   ##############################
    54   2.2566   ################################
    55   2.2566   ################################
    56   2.2892   ################################

Mean windowed entropy: 1.8956
Min entropy: 1.5628 at position 9 (most structured)
Max entropy: 2.2892 at position 56 (most varied)

Entropy spark: ▂▂▂▂▂▃▂▂▂▂▂▂▂▂▂▃▃▂▃▃▃▃▃▃▃▃▃▃▃▂▂▂▂▂▂▂▂▃▃▃▃▃▃▃▃▃▃▃▃

--- Methodological note ---
Windowed entropy is an exploratory visualization, not a statistical test.
Apparent patterns (dips and peaks) are expected from random variation in a
short sequence. Without a null model, no significance can be assigned to
specific regions. Interpret as descriptive, not inferential.
```

Mutual information: upper vs. lower trigram transitions

```
---
Mutual information: upper vs. lower trigram transitions
When two consecutive hexagrams differ, do the upper and lower trigrams change
independently, or is knowing that the upper trigram changed informative about
whether the lower trigram also changed? Mutual information quantifies this:
  0 bits = completely independent (knowing one tells you nothing about the other)
  higher = correlated (they tend to change together or stay together)
---
Joint distribution of trigram changes:
                       Lower unchanged  Lower changed
  Upper unchanged          0 (  0.0%)        4 (  6.3%)
```

```
  Upper changed            5 (  7.9%)        54 ( 85.7%)

Mutual information: 0.0078 bits
Normalized MI: 0.0230 (0=independent, 1=perfectly correlated)

Mean MI of random permutations: 0.0203 bits
King Wen percentile: 7.4%

--- Full 8-state trigram mutual information ---
Using actual trigram identities (8 possible values each) rather than
binary changed/unchanged, which preserves more information.
MI(upper, lower) across all 64 hexagrams: 0.000000 bits
(Expected: 0.0 — since all 64 combinations appear exactly once,
the trigrams are perfectly independent in the static set.)
Note: with all 64 upper/lower trigram combinations present exactly once,
independence is expected by construction (complete Latin square).
```

Yin-yang balance wave

```
---
Yin-yang balance wave
Each hexagram has 6 lines that are either yang (solid, 1) or yin (broken, 0).
Note: since the King Wen sequence contains all 64 possible hexagrams exactly
once, the total yin-yang count is necessarily balanced (192 each). This is a
mathematical tautology, not a property of the ordering. However, the LOCAL
balance as you move through the sequence IS a property of the ordering — it
shows where yang-dominant or yin-dominant hexagrams cluster together.
---
Total yang lines: 192 / 384 (50.0%)
Total yin lines:  192 / 384 (50.0%)
Mean yang per hexagram: 3.00

Running yang average (window=8):
 Pos  Yang  RunAvg     Visualization
   1     6    6.00   [                    |                   #]
   2     0    3.00   [                    #                    ]
   3     2    2.67   [                 #  |                    ]
   4     2    2.50   [                #   |                    ]
   5     4    2.80   [                  # |                    ]
   6     4    3.00   [                    #                    ]
   7     1    2.71   [                  # |                    ]
   8     1    2.50   [                #   |                    ]
   9     5    2.38   [               #    |                    ]
  10     5    3.00   [                    #                    ]
  11     3    3.12   [                    #                    ]
  12     3    3.25   [                    |#                   ]
  13     5    3.38   [                    | #                  ]
  14     5    3.50   [                    |  #                 ]
  15     1    3.50   [                    |  #                 ]
  16     1    3.50   [                    |  #                 ]
  17     3    3.25   [                    |#                   ]
  18     3    3.00   [                    #                    ]
  19     2    2.88   [                   #|                    ]
  20     2    2.75   [                  # |                    ]
  21     3    2.50   [                #   |                    ]
  22     3    2.25   [               #    |                    ]
  23     1    2.25   [               #    |                    ]
  24     1    2.25   [               #    |                    ]
  25     4    2.38   [               #    |                    ]
  26     4    2.50   [                #   |                    ]
  27     2    2.50   [                #   |                    ]
  28     4    2.75   [                  # |                    ]
  29     2    2.62   [                 #  |                    ]
  30     4    2.75   [                  # |                    ]
  31     3    3.00   [                    #                    ]
  32     3    3.25   [                    |#                   ]
  33     4    3.25   [                    |#                   ]
  34     4    3.25   [                    |#                   ]
  35     2    3.25   [                    |#                   ]
  36     2    3.00   [                    #                    ]
  37     4    3.25   [                    |#                   ]
```

```
  38     4    3.25   [                    |#                   ]
  39     2    3.12   [                    #                    ]
  40     2    3.00   [                    #                    ]
  41     3    2.88   [                   #|                    ]
  42     3    2.75   [                  # |                    ]
  43     5    3.12   [                    #                    ]
  44     5    3.50   [                    |  #                 ]
  45     2    3.25   [                    |#                   ]
  46     2    3.00   [                    #                    ]
  47     3    3.12   [                    #                    ]
  48     3    3.25   [                    |#                   ]
  49     4    3.38   [                    | #                  ]
  50     4    3.50   [                    |  #                 ]
  51     2    3.12   [                    #                    ]
  52     2    2.75   [                  # |                    ]
  53     3    2.88   [                   #|                    ]
  54     3    3.00   [                    #                    ]
  55     3    3.00   [                    #                    ]
  56     3    3.00   [                    #                    ]
  57     4    3.00   [                    #                    ]
  58     4    3.00   [                    #                    ]
  59     3    3.12   [                    #                    ]
  60     3    3.25   [                    |#                   ]
  61     4    3.38   [                    | #                  ]
  62     2    3.25   [                    |#                   ]
  63     3    3.25   [                    |#                   ]
  64     3    3.25   [                    |#                   ]

Yang count spark: █ ▂▂▅▅▁▁▅▅▂▂▅▅▁▁▃▃▂▂▃▃▁▁▅▅▂▅▂▅▃▃▅▅▂▂▅▅▂▂▃▃▆▆▂▂▃▃▅▅▂▂▃▃▃▃▅▅▃▃▅▂▃▃

--- Yang/Yin dominance runs ---
  Positions 02-04: 3 consecutive yin-dominant hexagrams
```

Odd-vs-even transition parity (McKenna 25/75)

```
---
Odd-vs-even transition parity (McKenna's 25/75 observation)
---
Each transition d(s_i, s_{i+1}) is between 1 and 6 line changes.
Theorem 1 says within-pair distances are always even (in {2,4,6}),
so all 32 within-pair transitions are forced even by C1. The 31
between-pair transitions can be any non-5 value.

McKenna (The Invisible Landscape, 1975) noted that ~25% of King Wen's
transitions are odd-distance and ~75% are even-distance. We report both
the LINEAR reading (the standard 63-transition difference wave) and the
CIRCULAR reading (64 transitions, including the wrap-around s_63 -> s_0).

Last hexagram (position 64): #42 (101010)
First hexagram (position 1): #63 (111111)
Wrap-around distance: hamming(#42, #63) = 3 (odd)

Mode                       Total    Odd       %Odd   Even      %Even
LINEAR (63 trans)             63     15   23.8095%     48   76.1905%
CIRCULAR (64 trans)           64     16   25.0000%     48   75.0000%

CIRCULAR parity is EXACTLY 25.0000% / 75.0000% — McKenna's 25/75 holds exactly
when the wrap-around transition is included. Without wrap-around, the linear
split is 23.8095% / 76.1905% — approximately but not exactly 25/75.

--- Structural note ---
All 32 within-pair transitions are forced even by C1 + Theorem 1.
So the parity split is determined entirely by the 31 (or 32 with wrap)
between-pair transitions:
  Between-pair transitions: 15 odd, 16 even
  Within-pair transitions: 0 odd (forced), 32 even (forced)
  Plus wrap-around: 1 odd

--- 'C8' candidate: is the wrap-around parity a structural constraint? ---
The spec's C1-C7 do not constrain the wrap-around s_63 -> s_0. So if
McKenna's exact 25/75 reflects a real constraint on the sequence rather
```

```
than a numerical coincidence, it would imply a new constraint:
  C8 (weak):   hamming(s_63, s_0) is odd  (admits {1, 3, 5}; it narrows to
               {1, 3} only if C2 is ALSO extended to the wrap, which C1-C7 do not)
  C8 (strict): hamming(s_63, s_0) = 3  (King Wen's exact value)

CORRECTED 2026-08-01. This block previously called the weak form restrictive,
estimating P(odd) from a uniform prior and concluding it eliminated ~60% of
completions. That was wrong, and it contradicted this repository's own proven
theorem (SPECIFICATION.md, 'Wrap-around parity is odd').

The weak form eliminates NOTHING: odd wrap-around parity is FORCED by C5, not
probable under it. Proof: popcount(x^y) = popcount(x) + popcount(y) (mod 2), so
summing over the 63 transitions telescopes -- every interior hexagram appears in
exactly two transitions and cancels -- leaving
  sum of the 63 transition distances = popcount(s_0) + popcount(s_63)   (mod 2).
C5 fixes that multiset as {1:2, 2:20, 3:13, 4:19, 6:9}, whose sum is 211, odd.
Hence hamming(s_63, s_0) = popcount(s_63 ^ s_0) is odd for EVERY C5-satisfying
sequence. P(odd) = 1, not the 0.40 previously printed here.

So the question is deductive, not empirical, and no Monte Carlo is needed. Only
the STRICT form carries information: constraining the wrap to d = 3 filters
~8.2% of canonical orderings (measured at the 560T canonical), not ~80%.
Neither form is in the formal spec: the weak one is a theorem and adds nothing,
and the strict one is King Wen's own value, hence data-like by construction.
```

Hexagram neighborhoods

```
---
Hexagram neighborhoods
For each hexagram, which others are 'nearby' (differ by only 1 or 2 lines)?
Dense neighborhoods mean a hexagram has many close relatives; sparse ones are
more isolated. This also shows whether King Wen places neighbors near each
other in the sequence or scatters them.
---
01 ䷀ Heaven over Heaven           d=1: [09, 10, 13, 14, 43, 44] (6 neighbors, mean seq dist: 21.2)
02 ䷁ Earth over Earth             d=1: [07, 08, 15, 16, 23, 24] (6 neighbors, mean seq dist: 13.5)
03 ䷂ Water over Thunder           d=1: [08, 17, 24, 42, 60, 63] (6 neighbors, mean seq dist: 32.7)
04 ䷃ Mountain over Water          d=1: [07, 18, 23, 41, 59, 64] (6 neighbors, mean seq dist: 31.3)
05 ䷄ Water over Heaven            d=1: [09, 11, 43, 48, 60, 63] (6 neighbors, mean seq dist: 34.0)
06 ䷅ Heaven over Water            d=1: [10, 12, 44, 47, 59, 64] (6 neighbors, mean seq dist: 33.3)
07 ䷆ Earth over Water             d=1: [02, 04, 19, 29, 40, 46] (6 neighbors, mean seq dist: 19.0)
08 ䷇ Water over Earth             d=1: [02, 03, 20, 29, 39, 45] (6 neighbors, mean seq dist: 18.7)
09 ䷈ Wind over Heaven             d=1: [01, 05, 26, 37, 57, 61] (6 neighbors, mean seq dist: 26.2)
10 ䷉ Heaven over Lake             d=1: [01, 06, 25, 38, 58, 61] (6 neighbors, mean seq dist: 25.8)
11 ䷊ Earth over Heaven            d=1: [05, 19, 26, 34, 36, 46] (6 neighbors, mean seq dist: 18.7)
12 ䷋ Heaven over Earth            d=1: [06, 20, 25, 33, 35, 45] (6 neighbors, mean seq dist: 17.3)
13 ䷌ Heaven over Fire             d=1: [01, 25, 30, 33, 37, 49] (6 neighbors, mean seq dist: 20.2)
14 ䷍ Fire over Heaven             d=1: [01, 26, 30, 34, 38, 50] (6 neighbors, mean seq dist: 20.2)
15 ䷎ Earth over Mountain          d=1: [02, 36, 39, 46, 52, 62] (6 neighbors, mean seq dist: 28.8)
16 ䷏ Thunder over Earth           d=1: [02, 35, 40, 45, 51, 62] (6 neighbors, mean seq dist: 27.8)
17 ䷐ Lake over Thunder            d=1: [03, 25, 45, 49, 51, 58] (6 neighbors, mean seq dist: 26.2)
18 ䷑ Mountain over Wind           d=1: [04, 26, 46, 50, 52, 57] (6 neighbors, mean seq dist: 25.8)
19 ䷒ Earth over Lake              d=1: [07, 11, 24, 41, 54, 60] (6 neighbors, mean seq dist: 20.5)
20 ䷓ Wind over Earth              d=1: [08, 12, 23, 42, 53, 59] (6 neighbors, mean seq dist: 19.5)
21 ䷔ Fire over Thunder            d=1: [25, 27, 30, 35, 38, 51] (6 neighbors, mean seq dist: 13.3)
22 ䷕ Mountain over Fire           d=1: [26, 27, 30, 36, 37, 52] (6 neighbors, mean seq dist: 12.7)
23 ䷖ Mountain over Earth          d=1: [02, 04, 20, 27, 35, 52] (6 neighbors, mean seq dist: 14.7)
24 ䷗ Earth over Thunder           d=1: [02, 03, 19, 27, 36, 51] (6 neighbors, mean seq dist: 15.0)
25 ䷘ Heaven over Thunder          d=1: [10, 12, 13, 17, 21, 42] (6 neighbors, mean seq dist: 11.5)
26 ䷙ Mountain over Heaven         d=1: [09, 11, 14, 18, 22, 41] (6 neighbors, mean seq dist: 11.8)
27 ䷚ Mountain over Thunder        d=1: [21, 22, 23, 24, 41, 42] (6 neighbors, mean seq dist: 7.8)
28 ䷛ Lake over Wind               d=1: [31, 32, 43, 44, 47, 48] (6 neighbors, mean seq dist: 12.8)
29 ䷜ Water over Water             d=1: [07, 08, 47, 48, 59, 60] (6 neighbors, mean seq dist: 23.5)
30 ䷝ Fire over Fire               d=1: [13, 14, 21, 22, 55, 56] (6 neighbors, mean seq dist: 16.8)
31 ䷞ Lake over Mountain           d=1: [28, 33, 39, 45, 49, 62] (6 neighbors, mean seq dist: 12.7)
32 ䷟ Thunder over Wind            d=1: [28, 34, 40, 46, 50, 62] (6 neighbors, mean seq dist: 12.7)
33 ䷠ Heaven over Mountain         d=1: [12, 13, 31, 44, 53, 56] (6 neighbors, mean seq dist: 16.2)
34 ䷡ Thunder over Heaven          d=1: [11, 14, 32, 43, 54, 55] (6 neighbors, mean seq dist: 15.8)
35 ䷢ Fire over Earth              d=1: [12, 16, 21, 23, 56, 64] (6 neighbors, mean seq dist: 19.7)
36 ䷣ Earth over Fire              d=1: [11, 15, 22, 24, 55, 63] (6 neighbors, mean seq dist: 19.7)
37 ䷤ Wind over Fire               d=1: [09, 13, 22, 42, 53, 63] (6 neighbors, mean seq dist: 19.0)
```

```
38 ䷥ Fire over Lake              d=1: [10, 14, 21, 41, 54, 64] (6 neighbors, mean seq dist: 19.0)
39 ䷦ Water over Mountain          d=1: [08, 15, 31, 48, 53, 63] (6 neighbors, mean seq dist: 18.3)
40 ䷧ Thunder over Water           d=1: [07, 16, 32, 47, 54, 64] (6 neighbors, mean seq dist: 18.3)
41 ䷨ Mountain over Lake           d=1: [04, 19, 26, 27, 38, 61] (6 neighbors, mean seq dist: 18.5)
42 ䷩ Wind over Thunder            d=1: [03, 20, 25, 27, 37, 61] (6 neighbors, mean seq dist: 19.5)
43 ䷪ Lake over Heaven             d=1: [01, 05, 28, 34, 49, 58] (6 neighbors, mean seq dist: 20.8)
44 ䷫ Heaven over Wind             d=1: [01, 06, 28, 33, 50, 57] (6 neighbors, mean seq dist: 21.2)
45 ䷬ Lake over Earth              d=1: [08, 12, 16, 17, 31, 47] (6 neighbors, mean seq dist: 23.8)
46 ䷭ Earth over Wind              d=1: [07, 11, 15, 18, 32, 48] (6 neighbors, mean seq dist: 24.8)
47 ䷮ Lake over Water              d=1: [06, 28, 29, 40, 45, 58] (6 neighbors, mean seq dist: 16.3)
48 ䷯ Water over Wind              d=1: [05, 28, 29, 39, 46, 57] (6 neighbors, mean seq dist: 17.0)
49 ䷰ Lake over Fire               d=1: [13, 17, 31, 43, 55, 63] (6 neighbors, mean seq dist: 18.7)
50 ䷱ Fire over Wind               d=1: [14, 18, 32, 44, 56, 64] (6 neighbors, mean seq dist: 18.7)
51 ䷲ Thunder over Thunder         d=1: [16, 17, 21, 24, 54, 55] (6 neighbors, mean seq dist: 22.2)
52 ䷳ Mountain over Mountain       d=1: [15, 18, 22, 23, 53, 56] (6 neighbors, mean seq dist: 22.5)
53 ䷴ Wind over Mountain           d=1: [20, 33, 37, 39, 52, 57] (6 neighbors, mean seq dist: 14.7)
54 ䷵ Thunder over Lake            d=1: [19, 34, 38, 40, 51, 58] (6 neighbors, mean seq dist: 15.3)
55 ䷶ Thunder over Fire            d=1: [30, 34, 36, 49, 51, 62] (6 neighbors, mean seq dist: 13.7)
56 ䷷ Fire over Mountain           d=1: [30, 33, 35, 50, 52, 62] (6 neighbors, mean seq dist: 14.3)
57 ䷸ Wind over Wind               d=1: [09, 18, 44, 48, 53, 59] (6 neighbors, mean seq dist: 19.2)
58 ䷹ Lake over Lake               d=1: [10, 17, 43, 47, 54, 60] (6 neighbors, mean seq dist: 20.2)
59 ䷺ Wind over Water              d=1: [04, 06, 20, 29, 57, 61] (6 neighbors, mean seq dist: 30.2)
60 ䷻ Water over Lake              d=1: [03, 05, 19, 29, 58, 61] (6 neighbors, mean seq dist: 31.2)
61 ䷼ Wind over Lake               d=1: [09, 10, 41, 42, 59, 60] (6 neighbors, mean seq dist: 24.2)
62 ䷽ Thunder over Mountain        d=1: [15, 16, 31, 32, 55, 56] (6 neighbors, mean seq dist: 27.8)
63 ䷾ Water over Fire              d=1: [03, 05, 36, 37, 39, 49] (6 neighbors, mean seq dist: 34.8)
64 ䷿ Fire over Water              d=1: [04, 06, 35, 38, 40, 50] (6 neighbors, mean seq dist: 35.2)

--- Neighborhood clustering in King Wen sequence ---
Mean sequence distance between Hamming-1 neighbors: 20.6
Mean sequence distance between any two hexagrams:   21.7
Hamming-1 neighbors are closer than average in the sequence (clustered).

--- Neighborhood clustering null model ---
Shuffling binary hexagrams 10,000 times to build a null distribution
of mean sequence distance between Hamming-1 neighbors.
Null distribution: min=17.9, mean=21.7, max=25.2
King Wen observed mean: 20.6
Percentile: 11.8% (proportion of shuffles with mean <= King Wen's)
King Wen's neighborhood clustering is within the range expected by chance.
```

Recurrence plot

```
---
Recurrence plot
A recurrence plot shows where the difference wave repeats itself. Each cell (i,j)
is marked if the difference at position i equals the difference at position j.
Diagonal lines indicate repeated sequences; vertical/horizontal lines indicate
values that persist. Clusters of marks reveal recurring local patterns.
---
              1         2         3         4         5         6
     123456789012345678901234567890123456789012345678901234567890123
  1  +.........#.....#.........#.#........#..............#.......#.#
  2  .+....#.#..##.##....###..#.#..#....#....#.#...#......##...#....
  3  ..+##..#.#...#....#.....#.......###.#.#..#..#...###.....#......
  4  ..#+#..#.#...#....#.....#.......###.#.#..#..#...###.....#......
  5  ..##+..#.#...#....#.....#.......###.#.#..#..#...###.....#......
  6  .....+...........#.#...#.....#.#.......#...#.#.#.......#.#...#.
  7  .#....+.#..##.##....###..#.#..#....#....#.#...#......##...#....
  8  ..###..+.#...#....#.....#.......###.#.#..#..#...###.....#......
  9  .#....#.+..##.##....###..#.#..#....#....#.#...#......##...#....
 10  ..###..#.+...#....#.....#.......###.#.#..#..#...###.....#......
 11  #.........+.....#.........#.#........#..............#.......#.#
 12  .#....#.#..+#.##....###..#.#..#....#....#.#...#......##...#....
 13  .#....#.#..#+.##....###..#.#..#....#....#.#...#......##...#....
 14  ..###..#.#...+....#.....#.......###.#.#..#..#...###.....#......
 15  .#....#.#..##.+#....###..#.#..#....#....#.#...#......##...#....
 16  .#....#.#..##.#+....###..#.#..#....#....#.#...#......##...#....
 17  #.........#.....+.........#.#........#..............#.......#.#
 18  .....#...........+.#...#.....#.#.......#...#.#.#.......#.#...#.
 19  ..###..#.#...#....+.....#.......###.#.#..#..#...###.....#......
```

```
 20  .....#...........#.+...#.....#.#.......#...#.#.#.......#.#...#.
 21  .#....#.#..##.##....+##..#.#..#....#....#.#...#......##...#....
 22  .#....#.#..##.##....#+#..#.#..#....#....#.#...#......##...#....
 23  .#....#.#..##.##....##+..#.#..#....#....#.#...#......##...#....
 24  .....#...........#.#...+.....#.#.......#...#.#.#.......#.#...#.
 25  ..###..#.#...#....#.....+.......###.#.#..#..#...###.....#......
 26  .#....#.#..##.##....###..+.#..#....#....#.#...#......##...#....
 27  #.........#.....#.........+.#........#..............#.......#.#
 28  .#....#.#..##.##....###..#.+..#....#....#.#...#......##...#....
 29  #.........#.....#.........#.+........#..............#.......#.#
 30  .....#...........#.#...#.....+.#.......#...#.#.#.......#.#...#.
 31  .#....#.#..##.##....###..#.#..+....#....#.#...#......##...#....
 32  .....#...........#.#...#.....#.+.......#...#.#.#.......#.#...#.
 33  ..###..#.#...#....#.....#.......+##.#.#..#..#...###.....#......
 34  ..###..#.#...#....#.....#.......#+#.#.#..#..#...###.....#......
 35  ..###..#.#...#....#.....#.......##+.#.#..#..#...###.....#......
 36  .#....#.#..##.##....###..#.#..#....+....#.#...#......##...#....
 37  ..###..#.#...#....#.....#.......###.+.#..#..#...###.....#......
 38  #.........#.....#.........#.#........+..............#.......#.#
 39  ..###..#.#...#....#.....#.......###.#.+..#..#...###.....#......
 40  .....#...........#.#...#.....#.#.......+...#.#.#.......#.#...#.
 41  .#....#.#..##.##....###..#.#..#....#....+.#...#......##...#....
 42  ..###..#.#...#....#.....#.......###.#.#..+..#...###.....#......
 43  .#....#.#..##.##....###..#.#..#....#....#.+...#......##...#....
 44  .....#...........#.#...#.....#.#.......#...+.#.#.......#.#...#.
 45  ..###..#.#...#....#.....#.......###.#.#..#..+...###.....#......
 46  .....#...........#.#...#.....#.#.......#...#.+.#.......#.#...#.
 47  .#....#.#..##.##....###..#.#..#....#....#.#...+......##...#....
 48  .....#...........#.#...#.....#.#.......#...#.#.+.......#.#...#.
 49  ..###..#.#...#....#.....#.......###.#.#..#..#...+##.....#......
 50  ..###..#.#...#....#.....#.......###.#.#..#..#...#+#.....#......
 51  ..###..#.#...#....#.....#.......###.#.#..#..#...##+.....#......
 52  ...................................................+.......#...
 53  #.........#.....#.........#.#........#..............+.......#.#
 54  .#....#.#..##.##....###..#.#..#....#....#.#...#......+#...#....
 55  .#....#.#..##.##....###..#.#..#....#....#.#...#......#+...#....
 56  .....#...........#.#...#.....#.#.......#...#.#.#.......+.#...#.
 57  ..###..#.#...#....#.....#.......###.#.#..#..#...###.....+......
 58  .....#...........#.#...#.....#.#.......#...#.#.#.......#.+...#.
 59  .#....#.#..##.##....###..#.#..#....#....#.#...#......##...+....
 60  ...................................................#.......+...
 61  #.........#.....#.........#.#........#..............#.......+.#
 62  .....#...........#.#...#.....#.#.......#...#.#.#.......#.#...+.
 63  #.........#.....#.........#.#........#..............#.......#.+

Recurrence rate: 952/3906 (24.4%)

--- Recurrence rate null model ---
Theoretical expected recurrence rate (sum of p_i^2): 25.6%

King Wen recurrence rate:       24.4%
Mean random recurrence rate:    23.3%
King Wen percentile vs random:  71.8%
```

DNA codon mapping

```
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DNA codon mapping
Both the I Ching (64 hexagrams) and the genetic code (64 codons) are systems
of 64 elements built from binary-like pairs: yin/yang for hexagrams, and the
base pairs A-T and G-C for DNA. Each codon is a sequence of 3 bases from the
set {A, C, G, U}, giving 4^3 = 64 possibilities. While this is a numerical
coincidence (both are 2^6 = 64), comparing their structural properties is
instructive about combinatorial systems in general.
Caveat: the mapping of bit pairs to DNA bases (00=U, 01=C, 10=A, 11=G) is one
of 24 possible assignments. Different mappings produce different codon assignments
and different degeneracy statistics. No mapping is privileged by either system.
This analysis is illustrative, not evidence of a biological connection.
---
Pos | Hex | Binary | Codon | Amino Acid | Name
----|-----|--------|-------|------------|----
```

```
01  | ䷀  | 111111 | GGG   | Gly        | Heaven over Heaven
02  | ䷁  | 000000 | UUU   | Phe        | Earth over Earth
03  | ䷂  | 010001 | CUC   | Leu        | Water over Thunder
04  | ䷃  | 100010 | AUA   | Ile        | Mountain over Water
05  | ䷄  | 010111 | CCG   | Pro        | Water over Heaven
06  | ䷅  | 111010 | GAA   | Glu        | Heaven over Water
07  | ䷆  | 000010 | UUA   | Leu        | Earth over Water
08  | ䷇  | 010000 | CUU   | Leu        | Water over Earth
09  | ䷈  | 110111 | GCG   | Ala        | Wind over Heaven
10  | ䷉  | 111011 | GAG   | Glu        | Heaven over Lake
11  | ䷊  | 000111 | UCG   | Ser        | Earth over Heaven
12  | ䷋  | 111000 | GAU   | Asp        | Heaven over Earth
13  | ䷌  | 111101 | GGC   | Gly        | Heaven over Fire
14  | ䷍  | 101111 | AGG   | Arg        | Fire over Heaven
15  | ䷎  | 000100 | UCU   | Ser        | Earth over Mountain
16  | ䷏  | 001000 | UAU   | Tyr        | Thunder over Earth
17  | ䷐  | 011001 | CAC   | His        | Lake over Thunder
18  | ䷑  | 100110 | ACA   | Thr        | Mountain over Wind
19  | ䷒  | 000011 | UUG   | Leu        | Earth over Lake
20  | ䷓  | 110000 | GUU   | Val        | Wind over Earth
21  | ䷔  | 101001 | AAC   | Asn        | Fire over Thunder
22  | ䷕  | 100101 | ACC   | Thr        | Mountain over Fire
23  | ䷖  | 100000 | AUU   | Ile        | Mountain over Earth
24  | ䷗  | 000001 | UUC   | Phe        | Earth over Thunder
25  | ䷘  | 111001 | GAC   | Asp        | Heaven over Thunder
26  | ䷙  | 100111 | ACG   | Thr        | Mountain over Heaven
27  | ䷚  | 100001 | AUC   | Ile        | Mountain over Thunder
28  | ䷛  | 011110 | CGA   | Arg        | Lake over Wind
29  | ䷜  | 010010 | CUA   | Leu        | Water over Water
30  | ䷝  | 101101 | AGC   | Ser        | Fire over Fire
31  | ䷞  | 011100 | CGU   | Arg        | Lake over Mountain
32  | ䷟  | 001110 | UGA   | Stop       | Thunder over Wind
33  | ䷠  | 111100 | GGU   | Gly        | Heaven over Mountain
34  | ䷡  | 001111 | UGG   | Trp        | Thunder over Heaven
35  | ䷢  | 101000 | AAU   | Asn        | Fire over Earth
36  | ䷣  | 000101 | UCC   | Ser        | Earth over Fire
37  | ䷤  | 110101 | GCC   | Ala        | Wind over Fire
38  | ䷥  | 101011 | AAG   | Lys        | Fire over Lake
39  | ䷦  | 010100 | CCU   | Pro        | Water over Mountain
40  | ䷧  | 001010 | UAA   | Stop       | Thunder over Water
41  | ䷨  | 100011 | AUG   | Met        | Mountain over Lake
42  | ䷩  | 110001 | GUC   | Val        | Wind over Thunder
43  | ䷪  | 011111 | CGG   | Arg        | Lake over Heaven
44  | ䷫  | 111110 | GGA   | Gly        | Heaven over Wind
45  | ䷬  | 011000 | CAU   | His        | Lake over Earth
46  | ䷭  | 000110 | UCA   | Ser        | Earth over Wind
47  | ䷮  | 011010 | CAA   | Gln        | Lake over Water
48  | ䷯  | 010110 | CCA   | Pro        | Water over Wind
49  | ䷰  | 011101 | CGC   | Arg        | Lake over Fire
50  | ䷱  | 101110 | AGA   | Arg        | Fire over Wind
51  | ䷲  | 001001 | UAC   | Tyr        | Thunder over Thunder
52  | ䷳  | 100100 | ACU   | Thr        | Mountain over Mountain
53  | ䷴  | 110100 | GCU   | Ala        | Wind over Mountain
54  | ䷵  | 001011 | UAG   | Stop       | Thunder over Lake
55  | ䷶  | 001101 | UGC   | Cys        | Thunder over Fire
56  | ䷷  | 101100 | AGU   | Ser        | Fire over Mountain
57  | ䷸  | 110110 | GCA   | Ala        | Wind over Wind
58  | ䷹  | 011011 | CAG   | Gln        | Lake over Lake
59  | ䷺  | 110010 | GUA   | Val        | Wind over Water
60  | ䷻  | 010011 | CUG   | Leu        | Water over Lake
61  | ䷼  | 110011 | GUG   | Val        | Wind over Lake
62  | ䷽  | 001100 | UGU   | Cys        | Thunder over Mountain
63  | ䷾  | 010101 | CCC   | Pro        | Water over Fire
64  | ䷿  | 101010 | AAA   | Lys        | Fire over Water

Unique amino acids in King Wen order: 21
Total possible amino acids + stop: 21

--- Degeneracy comparison ---
In DNA, many single-base changes preserve the amino acid (the code is 'degenerate').
Do single-line hexagram changes preserve the mapped amino acid?
Single-line changes that preserve amino acid: 100/384 (26.0%)
```

Shannon entropy analysis

```
---
Shannon entropy analysis
Entropy measures disorder: high entropy means the difference values are spread
evenly across all possibilities (random-looking), low entropy means certain values
dominate (structured). We compare the King Wen wave's entropy against thousands
of random permutations of the same 64 hexagrams. If King Wen's entropy is
unusually low, the sequence is more structured than random chance would produce.
---
King Wen difference wave entropy: 2.0759 bits
Maximum entropy (all 7 values): 2.8074 bits
Maximum entropy (5 observed values): 2.3219 bits
Mean entropy of random permutations: 2.1907 bits
Min random entropy observed: 1.7065 bits
Max random entropy observed: 2.4881 bits
King Wen percentile: 13.2% (lower = more structured)
Effect size (Cohen's d): -1.11 (negative = more structured than random)

--- Entropy conditioned on pair constraint ---
The unconstrained comparison above may be misleading: the pair structure
itself constrains the entropy. How does King Wen compare against random
orderings that also satisfy the pair constraint?
Mean pair-constrained entropy: 2.2349 bits
King Wen percentile (pair-constrained): 5.7%
(Similar to unconstrained percentile of 13.2%.)

--- Distribution comparison ---
Value    King Wen Expected (random avg)
  0            0                 0.0
  1            2                 6.1
  2           20                15.1
  3           13                19.7
  4           19                15.0
  5            0                 6.1
  6            9                 1.0
```

Path analysis (graph theory)

```
---
Path analysis (graph theory)
Imagine the 64 hexagrams as cities on a map, where the 'distance' between any
two is the number of lines that differ. The King Wen sequence is a route that
visits all 64 cities. Is it a short, efficient route (nearby hexagrams placed
next to each other), or a long, wandering one? We compare its total distance
against random routes and a greedy 'always go to the nearest unvisited city'
algorithm. This reveals whether the King Wen ordering was optimized for smooth
transitions or had other priorities.
---
King Wen total path length:  211 (sum of all line changes)
Greedy nearest-neighbor:     75
Mean random path length:     192.1
Min random observed:         156
Max random observed:         229
King Wen percentile:         97.6% (lower = shorter path)
Effect size (Cohen's d):     +2.02

--- Theoretical bounds ---
Minimum possible (63 transitions of 1): 63
Maximum possible (63 transitions of 6): 378
King Wen uses 55.8% of maximum path length

--- Pair-constrained comparison ---
The above compares against fully random orderings. A fairer comparison is
against random orderings that also preserve the pair structure (each pair
of hexagrams stays adjacent). This is the right null model for asking
whether King Wen's path length is unusual GIVEN its pair constraint.
Mean pair-constrained path length: 214.1
Min pair-constrained observed:     191
```

```
Max pair-constrained observed:     238
King Wen percentile (pair-constrained): 29.0%
Effect size (Cohen's d, pair-constrained): -0.47
```

Constraint satisfaction analysis

```
---
Constraint satisfaction analysis
The King Wen sequence satisfies several constraints simultaneously:
  1. All 32 consecutive pairs are either reverse or inverse (never unrelated)
  2. No consecutive hexagrams differ by exactly 5 lines
  3. No consecutive hexagrams are identical (0-line transitions)
How rare is it for a random ordering to satisfy all these at once? We test
random permutations against each constraint individually and combined.
---
Results from 10,000 random permutations:
  All pairs reverse/inverse:       0 (0.000%)
  No 5-line transitions:          13 (0.13%)
  Both constraints together:       0 (0.0000%)
  No random permutation satisfied both constraints.
  Statistical note: 0/10,000 gives a 95% upper bound of <0.0300%
  (i.e., the true rate is less than ~1 in 3,333, but we cannot
  distinguish 'impossible' from 'extremely rare' with this sample size).

--- Conditional probability: P(no-5 | pair constraint) ---
The pair structure constrains transitions within pairs (always even or 6),
so 5-line transitions can only occur at the 31 between-pair boundaries.
How often do pair-constrained orderings also avoid 5-line transitions?
  Pair-constrained trials: 100,000
  Also satisfy no-5:       4,361 (4.36%)
  Approximately 1 in 22 pair-constrained orderings avoid 5-line transitions.
  The no-5 property is uncommon but not extraordinary among pair-constrained orderings.

--- Sensitivity analysis: reversed bit convention ---
What if bit 0 = top line instead of bottom? All binary values reverse,
changing pair types and the difference wave. Key properties tested:
  Pair structure preserved:  Yes
  No-5 property preserved:   Yes
  Difference wave identical: Yes
  Original entropy: 2.0759, reversed: 2.0759
  Original path length: 211, reversed: 211
  All key properties are invariant under bit reversal.
```

Bootstrap confidence intervals

```
---
Bootstrap confidence intervals
The Monte Carlo results (e.g., '0.18% of random orderings avoid 5-line
transitions') are estimates based on sampling. Bootstrap resampling quantifies
how precise those estimates are by repeatedly resampling from the results and
computing confidence intervals. Narrower intervals = more reliable estimates.
---
Base trials: 10,000
Bootstrap resamples: 1000

No-5-line-transition rate: 0.150%
95% confidence interval: [0.080%, 0.230%]
Interval width: 0.150 percentage points

Approximately 1 in 667 random orderings
95% CI: 1 in 435 to 1 in 1250

Note: These CIs measure the precision of the Monte Carlo estimate, not
fundamental uncertainty about the true proportion. They reflect how much
the estimate would vary if you re-ran the simulation with 10,000 trials.
Increasing --trials narrows these CIs because the estimate becomes more precise.
```

Monte Carlo analysis

```
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Monte Carlo analysis (10,000 random permutations)
The King Wen sequence has a striking property: no two consecutive hexagrams
differ by exactly 5 lines. With 6 lines per hexagram and 7 possible difference
values (0-6), is avoiding 5 remarkable or just a coincidence? To find out, we
randomly shuffle the 64 hexagrams thousands of times and check how often a
random ordering also avoids 5-line transitions. The rarer it is, the less
plausible chance becomes as an explanation for the avoidance.
---
Permutations with no 5-line transitions: 17/10,000 (0.17%)
Approximately 1 in 588 random orderings share this property.
Odds ratio against random: 587:1
```